built the literature

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Causal Inference: The Remix · Yale University Press · forthcoming 2026

The National Supported Work Demonstration

A federally funded **randomized** job-training program.

1975–1979. Four target groups, including disadvantaged adult men.

Random assignment to treatment vs. control. Outcome: real earnings in 1978.

LaLonde's experimental benchmark

$$\widehat{\mathrm{ATT}}^{\mathrm{RCT}} \approx \$886$$

1978 dollars. Adult-male subsample.

Random assignment $\Rightarrow$ no selection. This is the truth.

Then LaLonde threw away the controls

He kept the treated — the 297 NSW men.

He replaced the controls with samples from the **CPS** and the **PSID**.

A simulation of what an applied researcher would face — no experiment, just observational data and methods.

Then he tried every method going at the time

- OLS with controls
- Two-step Heckman selection corrections
- Fixed effects on pre-program earnings
- Difference-in-differences

Could any of them recover $886?

No

Estimates ranged from $-$**\$16,000** to $+$**\$3,000**.

Most were negative. Most missed the benchmark by orders of magnitude.

LaLonde (1986), AER 76(4): 604–620.

The headline that shook applied econometrics

"The econometric procedures
cannot replicate the experimental results."

This is the paper that made program evaluation
move toward randomization and design.

Dehejia & Wahba reopened the case

Rajeev Dehejia and **Sadek Wahba**,
Review of Economics and Statistics (2002).

They asked: does **propensity score matching** do better than LaLonde's methods?

First, they re-cut the experimental sample

They restricted to NSW men with earnings observed in **both 1974 and 1975**.

A smaller, slightly different population.

This is the subsample the homework uses: $lalonde\_exp\_panel.dta$.

A new experimental benchmark

$$\widehat{\text{ATT}}^{\text{RCT}}_{\text{D\&W}} \approx \$1{,}794$$

Roughly **$1,700** in the way we usually report it.

Why bigger than LaLonde's $886? Different subsample, different ATT.

Why the headlines differ

- Both numbers come from the *same* NSW experiment.
- LaLonde used the full adult-male sample.
- D&W used a *subset* — only men with two years of pre-program earnings.
- Subsetting on a covariate changes the population whose ATT you're estimating.
- Same identification (random assignment). Different target group.

This is not a contradiction — it is the ATT defined on different units.

D&W’s quasi-experimental result

With **propensity-score matching** on CPS / PSID controls...

they recovered $\approx \$1{,}700$.

A constructive answer to LaLonde: design plus matching can work.

A puzzle

We used **non-experimental controls** (CPS, PSID)
in the larger LaLonde–D&W literature.

And yet the ATT we target is still $\approx \$1{,}700$.

Why doesn't the ATT change when we change the controls?

Because the ATT is a property of the treated

$$\text{ATT} = \mathbb{E}[Y^1 - Y^0 \mid D = 1, \text{ Post}]$$

Only the $D = 1$ side appears.

The control units are just a *tool* to learn about $\mathbb{E}[Y^0 \mid D{=}1, \text{Post}]$.

Random assignment gave us a clean tool

Under random assignment:

$$\mathbb{E}[Y^0 \mid D{=}1] \;=\; \mathbb{E}[Y^0 \mid D{=}0]$$

So $\mathbb{E}[Y \mid D{=}0, \text{Post}]$ *is* the counterfactual we need.

No selection bias.

Non-experimental controls break that equality

CPS men and NSW participants are not interchangeable.

$$\mathbb{E}[Y^0 \mid D{=}1] \;\neq\; \mathbb{E}[Y^0 \mid D{=}0]$$

Selection bias enters.

NSW men had worse pre-program earnings. The CPS pool didn't.

But the *target* hasn't changed

$\mathbb{E}[Y^0 \mid D{=}1, \text{Post}]$ is a feature of the NSW men.

It does not depend on who we picked as controls.

What the controls control is *our estimate* of that target —
and the size of the bias.

So what is DiD for, in this story?

- In the RCT sample: DiD just *confirms* the post-period difference-in-means.
- In the non-experimental sample: DiD tries to *net out* the pre-existing level gap between NSW and CPS men.
- The truth is fixed at $1,700. The question is whether DiD's parallel-trends assumption is enough to recover it.

Did it?

The benchmark and the challenge

RCT benchmark: $\$1,794$

Naive DiD on the non-experimental panel: $\$3,621$

We can do better than additive controls. Let's go to the code.

Throw away the experimental controls

Keep the 297 NSW treated men.

Replace the controls with a random sample of American households from the **CPS**.

Data: $lalonde\_nonexp\_panel.dta$. Same treated rows. New controls.

Selection is severe

	1974	1975	1978
NSW (treated, $n = 185$)	\$2,096	\$1,532	\$6,349
CPS (control, $n = 15{,}992$)	\$14,017	\$13,651	\$14,847
Diff	$-\$11{,}921$	$-\$12{,}119$	$-\$8{,}498$

$$\mathbb{E}[Y^0 \mid D{=}1] \ll \mathbb{E}[Y^0 \mid D{=}0]$$

The level gap is huge — and so is the growth-rate gap. DiD doesn't fix that.

The 2×2: where \$3,621 comes from

	1974	1975 (pre)	1978 (post)
NSW (treated, $n = 185$)	\$2,096	\$1,532	\$6,349
CPS (control, $n = 15{,}992$)	\$14,017	\$13,651	\$14,847

NSW change:	\$6,349 − \$1,532 =	\$4,817
CPS change:	\$14,847 − \$13,651 =	\$1,196
DiD	\$4,817 − \$1,196 =	$\boxed{\$3{,}621}$

$$\$3{,}621 \neq \$1{,}794$$

NSW men bounce back fast from a deep dip. CPS men don't. DiD attributes the entire gap in growth rates to the program.

Naive DiD on this sample

$$\widehat{\delta}_{\text{naive DiD}} = +\$3{,}621$$

Right sign. *Wrong magnitude.* The benchmark is $\$1{,}794$.

NSW men had unusually low pre-earnings ($\$1{,}532$) and bounced back fast.
CPS controls didn't. DiD attributes the gap in growth rates entirely to treatment.

Spec 0: Unconditional parallel trends, no covariates

Assumption. Naive parallel trends: $E[\Delta Y(0) \mid D{=}1] = E[\Delta Y(0) \mid D{=}0]$ unconditionally.

Run:

```
reg re i.post##i.ever_treated, robust
```

Compare to $1,794.

Spec A: Conditional PT, additive X

Assumption. Conditional parallel trends with one slope per covariate (pre = post).

Run:

```
global X "age␣agesq␣educ␣educsq␣marr␣nodegree␣black␣hisp␣re74␣re75␣u
reg re i.post##i.ever_treated $X, robust
```

Compare to $1,794.

Spec B: Conditional PT, Post $\times X$

Assumption. Conditional parallel trends, allowing X-effects to differ pre vs post. Use ## so the spec includes *both* the pre-period X effect and its post-period change.

Run:

```
reg re i.post##i.ever_treated ///
    i.post##c.age i.post##c.agesq i.post##c.educ i.post##c.educsq ///
    i.post##i.marr i.post##i.nodegree i.post##i.black i.post##i.hisp ///
    i.post##c.re74 i.post##c.re75 i.post##i.u74 i.post##i.u75, robust
```

Compare to $1,794. *Single-# forces pre-period X effect to zero — not what we want.*

Spec C: Saturated TWFE / OR (HIT 1997)

Assumption. Conditional parallel trends; impute $Y(0)$'s trend on controls only and average over treated.

Run — by hand:

```
reshape wide re, i(id) j(post)
gen dy = re1 - re0

reg dy $X if ever_treated == 0
predict dy_hat
gen delta_hat = dy - dy_hat if ever_treated == 1
summarize delta_hat if ever_treated == 1
```

Cross-check with `drdid`:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) reg
```

Compare to $1,794. *Two numbers should agree to the dollar.*

Spec D: IPW (Abadie 2005)

Assumption. Conditional PT; correct propensity score. Treated weighted by 1; controls weighted by $\hat{p}(X)/(1-\hat{p}(X))$.

Run — by hand:

```
logit ever_treated $X
predict phat, pr
quietly sum ever_treated
local pr_treat = r(mean)
gen w_ipw = (ever_treated - phat) / (1 - phat) / ‘pr_treat’
gen␣ipw_dy␣=␣w_ipw␣*␣dy
summarize␣ipw_dy
```

Cross-check with `drdid`:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) ipw
```

Compare to $1,794.

Spec E: DR-DiD (Sant'Anna & Zhao 2020)

Assumption. Conditional PT, AND *either* $\hat{\mu}_0$ or $\hat{p}(X)$ is correct (not both required).

Run — by hand (in wide-FD form, `phat`/`dy_hat` already computed):

```
gen dr_t = ever_treated * (dy - dy_hat) / `pr_treat'
gen dr_c = (1 - ever_treated) * (phat/(1-phat)) * (dy - dy_hat) / `pr_treat'
quietly sum dr_t
local mt = r(mean)
quietly sum dr_c
display "DR ATT = " %9.0fc `mt' - r(mean)
```

Cross-check with `drdid`:

```
drdid re $X, time(year) ivar(id) tr(ever_treated) dripw
```

Compare to $1,794.

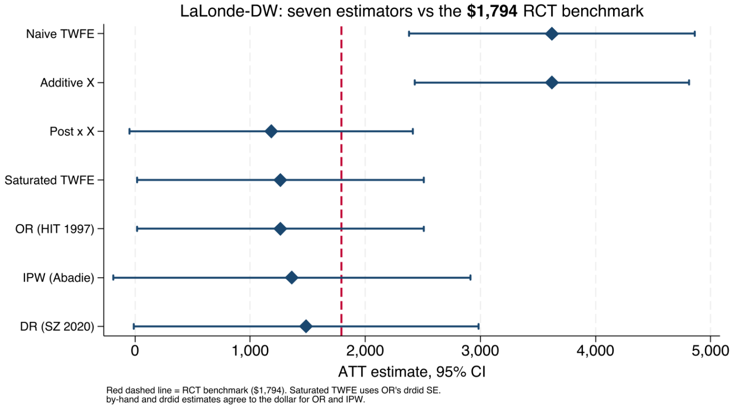
each one get?

The comparison

Spec	ATT	Distance from $1,794
0. Naive TWFE (no X)	$+\$3{,}621$	$+\$1{,}827$
A. Additive X	$+\$3{,}621$	$+\$1{,}827$
B. Post$\times X$ (## syntax)	$+\$1{,}184$	$-\$610$
C. Saturated TWFE (FD)	$+\$1{,}263$	$-\$531$
OR by hand $=$ `drdid reg`	$+\$1{,}263$	$-\$531$
D. IPW by hand $=$ `drdid ipw`	$+\$1{,}361$	$-\$433$
E. DR by hand	$+\$1{,}496$	$-\$298$
DR `drdid dripw`	$+\$1{,}485$	$-\$309$
RCT benchmark	$+\$1{,}794$	—

Naive $=$ Additive-X: the linear X control is absorbed. DR closest at $298 below. By-hand $=$ `drdid` to the dollar for OR and IPW; DR within $11.

Seven specs, one benchmark



Naive and Additive-X CIs sit entirely above $1,794. Every design-aware spec contains it.

DR lands closest; the others bracket it

- **DR** (\$1,496) sits **\$298 below** — closest observational estimator.
- **IPW** (\$1,361) sits \$433 below.
- **OR** / **Saturated TWFE** (\$1,263) sits \$531 below — two paths, same number.
- **Post**$\times X$ (\$1,184) sits \$610 below — still misses Bias_{HTE}.
- **Additive** X (\$3,621) doesn't move off Naive (\$3,621) — the FE absorbed the linear X.

"Doubly robust" = robust to one misspecification, not best in every sample. DR's 298 below is selection on unobservables — the residual no observational X can close.

What this lab teaches you

- The **ATT is fixed** ($\approx +\$1{,}794$). The controls only change the bias.
- **Additive X barely helps:** \$3,621 vs naive \$3,621. You need a design-aware estimator.
- Post$\times X$, OR, IPW, DR offer four different uses of the same X, all closer to truth.
- On this dataset, **DR-DiD lands closest** at \$298 below \$1,794.
- No method is magic. Each rests on an assumption you can name.

What additive X actually assumed

The additive-X spec $Y_{it} = \alpha + \delta(D_i \cdot \text{post}_t) + X_i'\beta + \varepsilon$ was a **TWFE specification choice**. Two restrictions came along free of charge:

1. **Homogeneous treatment effect in X.** One scalar δ. No room for $\delta(X)$.
2. **No time-varying X effect.** One β, used in both pre and post. If X's relationship with $Y(0)$ shifts between periods, the spec can't see it.

Neither restriction is identification. Both are specification.

Two TWFE relaxations

We saw both in the synthetic simulation. Both are still TWFE — just less restrictive parameterizations.

Post $\times X$. Let β differ pre vs post:

$$Y = \alpha_1 + \alpha_2 T + \alpha_3 D + \delta(DT) + \beta_1 X + \beta_2 XT + \varepsilon$$

Relaxes restriction (2). Still imposes (1): one scalar δ.

Fully saturated. Let β differ pre/post *and* let δ depend on X:

$$Y = \alpha_1 + \alpha_2 T + \alpha_3 D + \delta(DT) + \beta_1 X + \beta_2 XT + \beta_3 XD + \beta_4 DTX + \varepsilon$$

Relaxes both. The ATT is now $\delta + \beta_4\,\mathbb{E}[X \mid D{=}1]$ — average the X-specific effect over the treated.

The punchline

Additive X failed on LaLonde

not because the data are weird,

but because the **TWFE specification** was too restrictive.

Fully saturated TWFE = the regression form of HIT 1997 outcome regression.

Same estimator. Same ATT. Different machinery.

Spec C (FD with $D \times X$ interactions) and OR-by-hand returned the same dollar: \$1,263.

whether your method
can **see it.**